

MeterLink™ — BT's new telemetry service for meter reading

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MeterLink™ is a new BT service to read utility meters (water, electricity, gas). The BT network has been modified to give telemetry capability without affecting normal telephone service by the use of 'no-ring calls'. Accessed through the telemetry platform, no ring calls allow data to be communicated to and from customers' premises while the line is otherwise not in use.

The main focus of this paper is on the telemetry platform which uses the interactive speech applications platform (ISAP) and the telemetry interface unit which terminates telemetry calls at a customer's meter location.

1. Introduction

MeterLink™ is a new BT service for reading utility meters (electric, gas, water) over the public switched telephone network (PSTN). It is the first of a number of potential applications of network telemetry [1], transparently sharing telephone lines to collect and deliver relatively small amounts of data without affecting normal telephone calls. This paper will highlight the network elements of the underlying BT telemetry service and the role of the interactive speech applications platform (ISAP) as a PSTN calling engine and speech platform for the service.

This service is aimed at the 70 or so privatised utility companies who are experiencing new competitive and regulatory pressures. Electricity consumers of over 1 MW on a site (about 5000 sites nationwide) have been able to choose their supplier of both electricity and metering since 1990. From April 1994 this was extended to consumers of over 100 kW/site (about 50 000 nationwide) and by 1998 it is expected that electricity will become an open market with all consumers free to choose their supplier.

A similar situation is emerging in the gas market, where large gas consumers are already purchasing gas via the British Gas transport network. Competition in water supply is less developed and the regulatory focus to date has been on improving the quality of service while containing costs. However, the water industry is under considerable political pressure to introduce metering to domestic customers, which has heightened interest in telemetry.

For the utilities, remote meter reading is only attractive if it can save costs over the current mainly manual methods. Efficient processes have been developed whereby consumers take some readings and others are estimated so that the cost of manual reading is low. The utilities are, however, coming under regulatory pressure to read more accurately and more often. MeterLink has been designed to be cost-competitive with manual reading — where more actual readings are required the costs become even more attractive.

For the utility regulator, the BT telemetry service offers a very effective means of facilitating competition. Suitably equipped meters can be read throughout the UK, unlike broadcast telemetry systems which are limited in geographic coverage. Security is built into the system so that the utilities can only read meters to which access rights are owned. This facility is intended to preclude anti-competitive behaviour.

For BT, telemetry is a way of using the network when it might otherwise be idle, especially overnight. MeterLink is but one of a number of services which will be developed using this platform.

2. Systems architecture and major components

The technical requirements of the telemetry system, simply stated are to:

- automatically originate large numbers of calls per day (there are 23M households in the UK and many with two or three meters each to be read at least quarterly),
- route these calls to the appropriate location, regardless of any network diversions the telephone user may require,
- connect to and transfer data with a meter without ringing the consumer's telephone or interfering with normal telephone service ('no-ring calls'),
- route data back to the utility/MeterLink customer,
- charge for the service.

In addition, the utilities as well as BT are subject to regulatory controls and data must be segregated for data protection and competition reasons. BT must also consider the impact on the network of originating many calls automatically and build in safeguards so that the service will not cause congestion for normal calls. In the future it will be necessary to deliver telemetry over other networks including ISDN and other licensed operator networks.

These factors among others led to the systems architecture illustrated in Fig 1.

The functions of the systems elements in Fig 1 are given below.

- The telemetry interface system (TIS) [2] is a computer which is configurable to interface with the utility's existing systems and gives control of the service.
- The telemetry service management system holds data for the service and initiates call requests to the telemetry platform.
- Human assistance for users of the service is given by the telemetry service centre, which has terminals to monitor and operate the service.
- The telemetry platform is based on the interactive speech applications platform (ISAP) and is the calling engine for the whole telemetry service. Multiple telemetry platforms will be deployed as the service grows which will allow for very large numbers of calls; different types will also allow for other transport technologies.

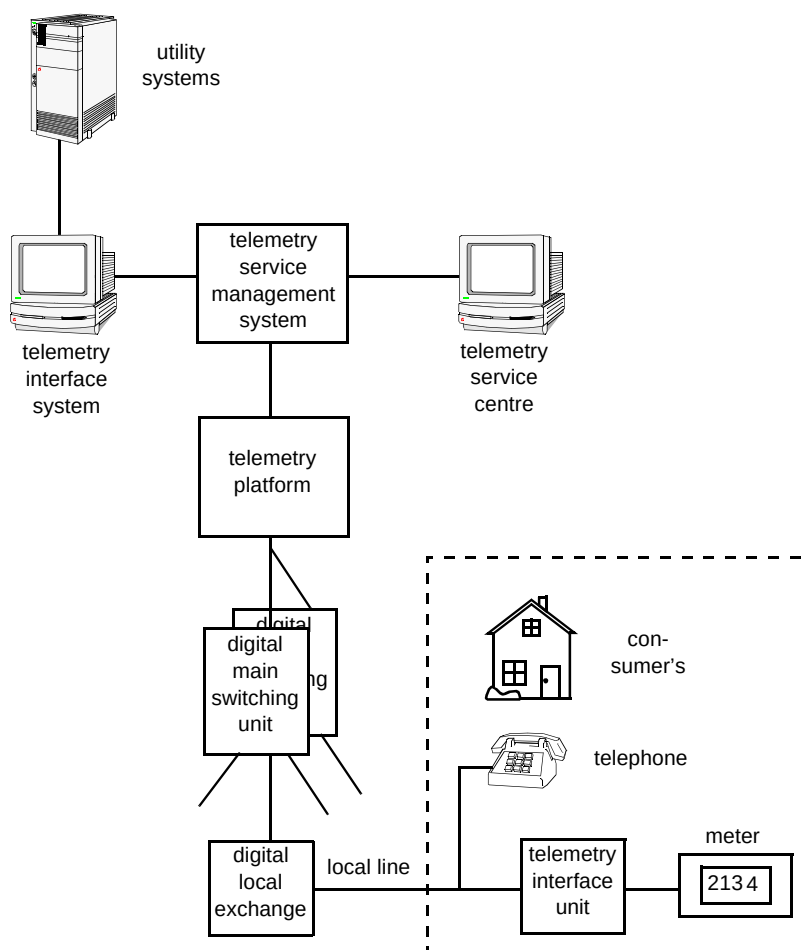


Fig 1 MeterLink systems architecture.

- Calls are routed into the network via a digital main switching unit (DMSU) which accepts CCITT SSC7 signalling and routes the calls from the telemetry platform. Two connections are used into the DMSU for resilience. Calls are then routed via digital local exchanges which have been modified to accept 'no-ring calls'.
- 'No-ring calls' cause tones to be sent to the customer's line but without the ringing which alerts the telephone customer to an incoming call. These tones instead activate a telemetry interface unit [3] which answers the call and then interfaces data between meters connected to the interface unit and the telemetry platform.

3. System functions

The MeterLink service features the following high-level functions.

3.1 Bulk read

Regular meter readings will be most efficiently made in bulk and the utility has the facility of creating lists of meters which can be read on a regular cycle. Lists can be created on the telemetry interface system and downloaded to the telemetry service management system for regular reading — the customer will normally consult telemetry service centre people to ensure that the correct priority is attached and terms of the contract agreed.

Lists of meters to be read will normally be downloaded on to the telemetry platform only when needed and will have a time window attached within which all readings should be made. The call request handler can then prioritise calls in order to give the best match of demand to resources and begin making calls on the list starting at the specified time.

Each call is associated with low-level network routing data and passed through the call rate control before handling by the call-connection/data-handling process. This makes 'no-ring calls' to the required telemetry interface unit and handles the resulting modem data transaction according to rules stored in the template for this kind of connection. Templates will have parameters passed and will translate data to a required format — a future upgrade will allow scripts to be run within the template so that intelligent handling of the transaction becomes possible.

Data is verified through a checksum and the information passed back by the telemetry platform, or an error code if the transaction fails (this will normally result from the telemetry service backing off to allow normal use of the line and need not indicate a fault condition). The call request handler is capable of retrying calls to a defined sequence (e.g. call two nights and if still no connection try one daytime call, useful for lines tied up overnight). The system

can also automatically try alternative strategies for initiating the 'no-ring call' including higher level wake up tones and preceding the call with line reversal (which may cause bell tinkle and so reverts to normal mode if possible).

3.2 On-line reading

The system may also be required to make a single reading as soon as possible, e.g. for home moves. The call sequence is exactly the same as above except that the highest priority is attached to the call. The call may still fail because the line is busy, but this information is passed back to the telemetry interface system so that the operator is informed of the reason why the meter cannot be read. It should also be noted that service management of the line during a house move has been considered in the design — changes in BT data will be co-ordinated so that contact with the meter is not lost in this transitional period.

3.3 Broadcast

A broadcast message can be initiated in much the same way as a bulk read, i.e. a list of meters to be contacted and each sent the same message via a template. The essential difference is that calls must be made in rapid sequence and might be initiated from more than one telemetry platform in parallel. This emulates the facility of broadcast systems to contact a number of meters in a very short period and has the added advantage that the broadcast message can be confirmed. For example, this feature will be especially useful if the broadcast is used to download a new tariff to the meter.

4. Service management

As a general principle, it should not be necessary for the utility to know the telephone number of the line used to deliver telemetry to its consumer. Since telemetry does not affect normal telephone use of the line, it is possible to use any line conveniently close to the meter for telemetry. This line could give telephone service to someone quite unrelated to the meter user, or could be ex-directory. Utilities will have their own reference for consumers which must be mapped across to the appropriate telephone number by the MeterLink service. The number mapping must then be maintained against telephone users/consumers moving house, changing network operator or simply going ex-directory.

The platform required for this function is the core of the service management system. Its primary function is that of holding consumer details and mapping these on to routing data for use by the network systems as required. Changes to consumer data are handled by links to BT's customer

service systems (CSS), which have themselves been modified to handle telemetry. The telemetry service management system is also the interface to the BT telemetry service via the telemetry platform, issuing requests for calls to/from meters and handling the data.

Utilities will control the MeterLink service through a telemetry interface system. This interface is customisable by systems integrators to fit in with the utilities' operations. An open interface to the telemetry service is specified (SIN252) to facilitate use of the service.

The service also has human operators who can help the utility to set up and use the service, via systems very similar to the telemetry interface system.

Utilities wishing to add meters to the service will be required to provide (electronically via the telemetry interface system, where possible) lists of consumer addresses for the MeterLink service which will be matched with BT data to ensure that there is a suitable line on the premises. Where a line is not present on the premises it may be possible to use an adjacent line, in which case a site survey will be needed. However, the bulk of installations will be processed automatically through normal line records. The telemetry service management system will then return lists of consumers which are pre-enrolled ready for installation of meters.

BT is not planning to provide and fit meters, but has instead provided automatic enrolment processes so that the fitters chosen by the utility can easily enrol meters themselves. This involves entering details about the specific meter and interface unit fitted on the premises, checked against the calling line identification of the line, so that the service can use the correct protocol and can check that these details match on later readings.

Once successfully enrolled and a check reading taken, utilities will be able to initiate meter readings through the TIS as and when required.

5. Telemetry platform

5.1 Why choose ISAP as the platform?

The telemetry platform is the core calling engine for telemetry services. Its requirements for PSTN use are that it:

- is capable of safe attachment to the core BT network (i.e. without risk of degrading core network call transport),

- has many modem channels for parallel calls to give a high calling rate,
- has modems capable of upgrade for emerging customer requirements,
- is expandable, reproducible (multiple platforms will be needed to give a scalable service),
- is capable of voice dialogue for enrolment (also permits other services).

Initial design studies in 1993 considered a number of options for this data application. These studies showed that the highly flexible digital signal processing (DSP) capabilities of the ISAP platform [4] were ideal for PSTN telemetry. The platform also had many other strong points:

- each speech shelf can handle a large number of modems, e.g. 108 V23 modems, and many shelves can be incorporated into a single platform giving the capability of incremental growth as the service takes off,
- powerful multi-processor UNIX environment, required for the many internal look-up and call-processing functions to proceed while maximising the use of DSP resources,
- reuse of core functionality of the ISAP to increase development efficiency and effectiveness (reduced risk over an entirely new platform),
- network hardened and certified (comprehensively tested to ensure that it cannot damage the network),
- network management capability,
- speech capability inherent, which gave the opportunity to automate the enrolment process and reduce the cost of meter installation.

The platform is also used for other applications, notably CallMinder, so that it will be familiar to operations personnel. These reasons led naturally to the choice of ISAP as the preferred platform for PSTN telemetry.

5.2 Internal operations of the telemetry platform

A simplified view of the internal operations of the telemetry platform is given in Fig 2.

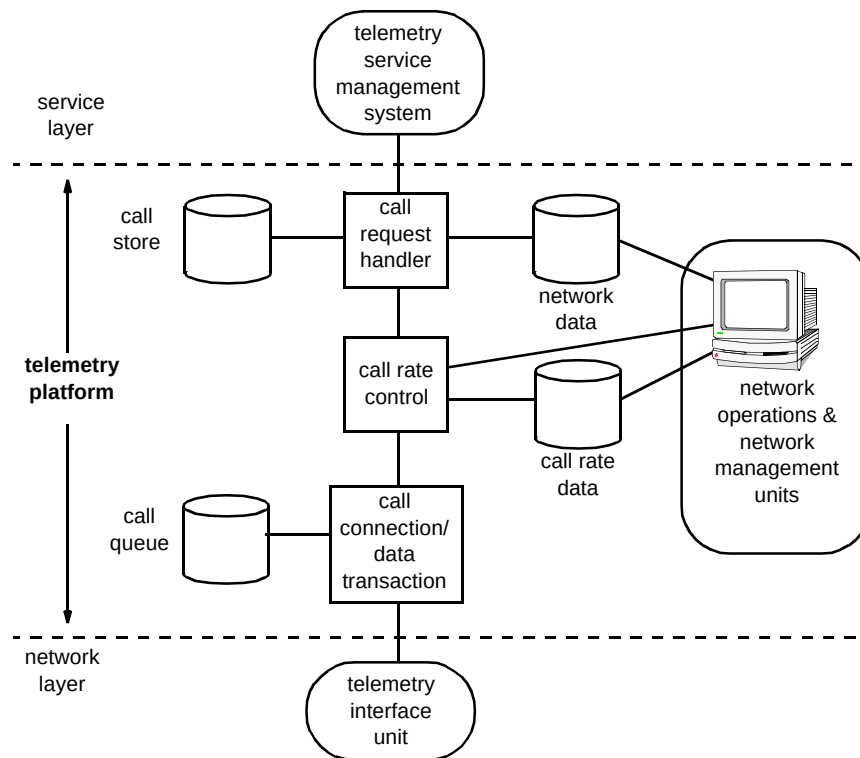


Fig 2 Telemetry application architecture.

Calls are received from the telemetry service management system and held in a queue. They will have different priorities depending on the service required but the system periodically increases the priority of calls so that the lowest priority calls will get handled after a time.

The generation of calls is carefully controlled through the calling rate control function [5]. Currently the telemetry platform is capable of originating 6000 calls/hour and this is expected to increase substantially as the software is optimised and resources added. Large batches of calls will become due at specific times of day or night at times specified by the utilities and unless managed by this function would cause step increases in the numbers of calls originated in the network. As this is not a normal calling pattern, it would normally register as a fault condition, in which case protection algorithms in the network would prevent these calls from being forwarded. Instead the call rate is increased at a controlled rate which avoids this possibility but nevertheless achieves the full calling rate in a relatively short time, e.g. 0 — 6000 calls/hour in less than 13 sec.

The calling rate control function also gives network operations personnel full control of the platform to protect the network under unusual circumstances, e.g. faults elsewhere in the network or unusually high calling patterns in the vicinity. Both fine control through manual setting of calling rates and coarse control through call gapping (setting a minimum time between call attempts) is given via secure links.

Routing telemetry calls to the required line, regardless of any diversions or network services in use, requires a form of sub-addressing within the digital local exchange. This requires a translation of the consumer's telephone number to a host network address. Because of the low-level nature of this address (to a specific concentrator and line card) the address length is variable and the number itself subject to change with engineering work on the network. A table mapping telephone numbers to host network addresses is held within the telemetry platform (network data in Fig 2) and maintained by network operations personnel.

Call connection/data transaction is via the bank of DSP modems which forms the heart of the telemetry platform [6]. Here a scheme of templates [7] is employed to translate high-level metering instructions to the protocol required by a specific type of meter and telemetry interface unit [8, 9]. This allows the telemetry interface system to request calls in a common format regardless of the equipment installed. Information held on the telemetry service management system invokes the appropriate template run on the telemetry platform to communicate with the installed meter and telemetry interface unit and return results in a format suitable for use by the telemetry interface system. This information is entered and stored automatically during the enrolment process.

Meter readings are only made for calls where the telemetry interface unit reference numbers (and optional security) match those as enrolled by the utility and are returned only to that utility. Information does not remain on

the telemetry platform for anyone else to read, although call requests can be retained for regular reading. Calls are returned to the telemetry service management system as soon as available.

6. Network changes required

6.1 Network infrastructure and signalling changes

The network infrastructure requirements for telemetry were considered along with the wider upgrade of the UK network needed for calling line identification (CLI). CLI requires a limited signalling facility in all digital local exchanges in order to pass calling numbers to customers' telephones. In the UK implementation of CLI this information is passed before any ringing is applied to line, i.e. as a 'no-ring call'. One benefit of this is that it can facilitate automatic routing of the call on customers' premises in accordance with the CLI number.

CLI is an embedded feature in the digital local exchanges which support it, and required upgrade of all line cards to include hardware for 'no-ring calls'. Accessing this feature for the telemetry service requires low-level access to this hardware and is via the host network address, as described in section 5.2.

Several classes of 'no-ring call' are available. The preferred type uses the line in entirely the idle state and the meter interface unit operates at a current level below that which will seize the line. This allows the most robust scheme for detecting that the telephone customer wants to initiate a call because looping the line is easily detected and the telemetry service can back-off. However, there will be circumstances where more line current is required and a loop-answer mode is permitted where the meter interface unit may take normal telephone line current. In this case both the telemetry platform and meter interface unit must continuously monitor line signal levels and drop the telemetry call if the telephone user lifts a handset.

Each 'no-ring call' is initiated with a wake-up tone which indicates the class of service required (different tones are used for CLI). Experience in the US with their simpler form of CLASS signalling has shown that it is less robust and more prone to false triggering. The BT system uses a dual wake-up tone with careful selection of the frequencies to avoid duplication of signalling tones used worldwide. CLASS signalling also normally includes a line reversal, which is easy to detect and confirm on the tone. This, however, can give rise to bell-tinkle, normally masked when ringing starts after the CLI is received. In the case of telemetry there is no following ringing and the line is likely to be used at night; the preferred mode of operation is therefore without line reversal to avoid bell tinkle where possible.

Before call set-up can be completed the exchange must detect that the local connection has been successfully made. This is indicated by returning a DTMF¹ 'C' tone to the exchange, which can then complete the connection. Loop termination 'no-ring calls' are of course detected by loop-answer as for normal calls. The 'no-ring call' is then connected through to the telemetry platform.

The types of 'no-ring call' are listed in Table 1.

Table 1 Types of 'no-ring call'.

No line reversal	Line reversal	
1	2	idle line
3	4	loop answer

6.2 Effects of MeterLink on the network

Telemetry services provide both an opportunity but also a potential problem for the UK PSTN. If the service is adopted widely for domestic metering, then very large call volumes are anticipated. Call volume will grow, especially if, as intended, other applications are added. These calls could swamp the PSTN if they are concentrated in time or location such as to exceed network capacity locally (the network as a whole has ample capacity for any foreseeable eventuality). The telemetry platform could also conceivably develop faults (for example continuous looping through a batch to generate erroneously high call rates). For these reasons a number of measures have to be included to protect the network against excess network loading.

The first level of control is to work with customers and set tariffs which encourage use of the network in less busy times. Figure 3 shows a typical calling pattern where the network is busy for large parts of the day and (increasingly) in the early evening when social calls are made and prices are lower.

Telemetry can use spare capacity overnight and for periods during the day — there is also always capacity for reasonable numbers of calls at any time for immediate meter reading.

¹ Dual tone multi-frequency signalling, as used in modern telephones. A matrix of four by four tones is defined, used two at a time to give the normal keypad 0—9 and *, # keys. Four combinations are not used on the keypad but are available for CLASS and telemetry signalling, referred to as 'A' through 'D'.

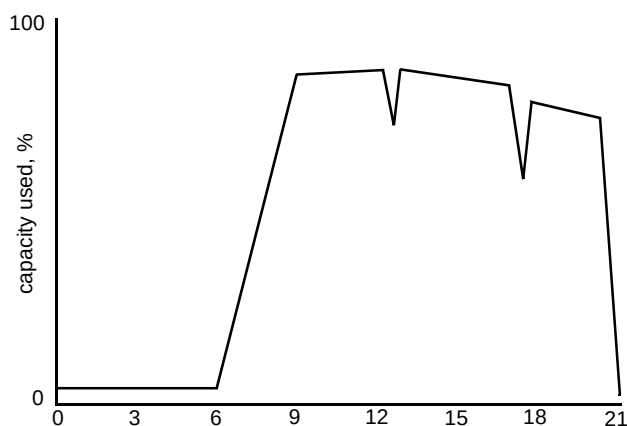


Fig 3 Calls in network (indicative).

Where it is necessary to place a large number of calls at the same time, e.g. a broadcast to meters in a town, the application will scatter calls among the list of host network addresses to be used to avoid overloading single points in the network.

As the network evolves it will also be possible to scatter calls between telemetry platforms to avoid overloading single routes. In exceptional circumstances, for instance, television programme response which can generate huge amounts of calls in a short period, it will be necessary for networks operations experts to set call controls on the telemetry service to preserve the PSTN performance; a measure of control is also built in for automatic operation to cover unforeseen and fault conditions.

Managing these risks has been an important consideration from the beginning of this development. In order to ensure that the required high standards are met, all network-affecting elements of the overall system have been tested and the telemetry platform and application subjected to particularly thorough testing, including provocative testing (attempts to induce network-affecting faults by, for example, removing live links/cards). While this is a tough regime within which to develop applications, the contribution of testing, has been highly constructive and networks test specialists continue to support the service during the current phase of rapid evolution to help ensure that the network certification given to MeterLink is maintained.

7. Telemetry in the consumer's premises

7.1 Connection to the telephone line

In order to receive telemetry calls a telemetry interface unit must be installed where it has access to both a telephone line and the utility meter. Normally the TIU will be

connected to the consumer's own telephone line; however, since the service is transparent to telephone use, this is not necessary and any convenient and suitable line will do.

Access to the line is via the standard line socket allowing telemetry interface units to be supplied and fitted by any organisation which can meet BABT requirements. The equipment will therefore be the property of the installer, but has the potential to disrupt service to the telephone user should a fault develop. Connecting via a line socket gives the user the ability to disconnect the telemetry interface unit under these circumstances.

It should be noted that this facility to disconnect the telemetry interface unit did cause some concern to the design team since this might allow some consumers to try to frustrate the meter reading service. However, advice from the utility experts indicated that a hard-wired connection from the line to the meter interface unit would be no more reliable since the type of customer who would frustrate the service is perfectly capable of cutting the wire.

7.2 Telemetry interface unit

The telemetry interface unit (see Fig 4) has the function of accepting telemetry 'no-ring calls' and converting meter readings into a form suitable for transmission over the network. The preferred mode of operation is the 'on-hook' mode where the TIU takes less than 2.5 mA in operation (less than 30 μ A on standby).

BT has developed a TIU [10] as proof that line-powered operation is feasible and to expedite early roll-out; however, the service has built-in flexibility for other modem standards and message protocols. The BT TIU will be used as an example for the rest of this section.

The unit is connected to the standard telephone line jack unit. The low standby current taken means that it will not affect the use of other telephone equipment in the house and does not reduce the total ringer equivalence number (REN) available to the telephone customer (REN is a measure of the loading of the line by telephones especially for ringing — up to a REN of four is allowed per line and the telephone user should not fit telephones in excess of this total).

The line interface unit contains all telephone termination and powering functions and is required to meet the provisions of BS 6319, etc, for connection to the UK network. It also contains a modem capable of V.23 signalling, half duplex 1200 bit/s in both directions plus 390 Hz back-channel carrier (as used in the V.23 standard, required for parallel off-hook detection when the TIU is used in off-hook mode). The service requires an MF generator for signalling call set-up, and wake-up tone detection.

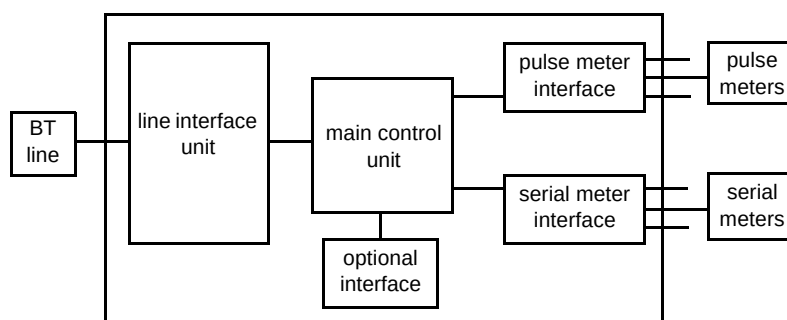


Fig 4 BT telemetry interface unit.

The wake-up tone has been specially selected to avoid false triggering. A guard tone of 1827 Hz is used for all calls, coupled with select tones of 520.6, 578.4 and 468 Hz for addressing up to three TIUs on the same line (more could be defined later). The select tone is cadenced 300 ms ON, 100 ms ON and 100 ms OFF to select the off-hook mode. This can be used if transmission proves unreliable for the on-hook mode and will be selected by the telemetry platform automatically if service on-hook is not reliable.

The main control unit contains a simple microprocessor and memory for supporting the message protocols required by the service. While the service is adaptable to any protocol, it is recommended that the RITA [11] protocol is used — this has been created in such a way that it makes efficient use of simple microprocessor memory mapping and is easy to monitor using simple protocol analysers.

Interface ports are provided for several pulse or serial interface meters. Pulse inputs simply count the number of contact closures, which gives a measure of incremental metering. The serial interface supports the IEC FLAG protocol, which is in wide use on electricity meters and has the benefits of standardisation and inherent safety isolation through the infra-red interface used.

Other interfaces are provided for a hand-held terminal (to ease set-up on installation) and a display unit which can optionally give the customer confirmation of the meter reading taken.

8. Service deployment

MeterLink is currently being trialled by major gas, regional electricity and water utilities. Even at the current early stage of installation many thousands of telemetry calls are made daily.

Currently, two telemetry platforms are deployed (for resilience) which gives sufficient capacity for the first four years of expected growth. The systems architecture has flexibility to increase ISAPs in line with business growth and share other ISAPs, e.g. CallMinder, when these have spare capacity, say at night.

The BT-designed telemetry interface units have been BABT approved for limited production and nearly 200 are operational in the Doncaster area. These units have proved reliable in use and have been of great value in testing the system under real operational conditions. Interface units have also been produced by GPT Ltd for major gas users and a number have been deployed to support trial MeterLink services. Other manufacturers have expressed an interest and are expected to supply interface units for telemetry.

9. Conclusions

This paper has given a top-level description of the systems which are needed to deliver BT telemetry and the MeterLink service. The telemetry platform has featured as an example of an application of the interactive speech applications platform which is concerned with data as well as speech.

The ISAP platform has proved to be highly capable of configuration for this challenging application, generating large numbers of calls with the flexibility to change the data format for each call. The platform also made it easy to add other features such as enrolment.

This flexibility will be essential as the telemetry service evolves and new applications are developed. For instance, telemetry will be able to communicate with vending machines to check on stock levels and notify the operator when these need restocking.

Acknowledgements

The telemetry service has required the development of highly complex systems in a very short time — driven by regulatory change.

We would like to thank our colleagues in BT/Networks and Systems for their help in developing this ISAP application.

The service builds on an early trial of automatic meter reading which used equipment supplied by Schlumberger and was run in collaboration with Yorkshire Electric and

Yorkshire Water companies. These customers have helped us a great deal in understanding the issues facing their industries and have been of enormous practical value in developing the service.

We have also enjoyed the support of many people across the whole BT group; in particular, the help of our colleagues in BT/Network Operations has been essential in ensuring we can deliver a reliable and manageable service.

References

- 1 SIN236 — telemetry service description (a copy can be obtained by calling 0800 318 601).
- 2 SIN252 — telemetry interface system — TSMS interface specification (a copy can be obtained by calling 0800 318 601).
- 3 SIN237 — telemetry CPE design guide (a copy can be obtained by calling 0800 318 601).
- 4 K R Rose and P M Hughes: 'Speech systems', BT Technol J, 13, No 3, pp 51 — 63 (April 1995).
- 5 D Tricker, G Ross and M Duffy: 'ISAP telemetry application architecture design', BT Document Reference 0106-0305-3-00004 (CMC).
- 6 S Lincoln and D Cleland: 'ISAP telemetry application physical design', BT Document Reference 0106-0390-3-00006 (CMC).
- 7 D Tricker: 'TIU transaction templates', BT Document Reference 0106-0305-3-00014 (CMC).
- 8 IEC 1107 Revision 1, 4th draft April 1994: 'Data exchange for meter reading, tariff and load control, direct local data exchange' (1994).
- 9 ISO 3309 — 07-01 2nd edition 'Data communications — high level data link control procedures — frame structure' (1979).
- 10 K E Nolde: 'Technical specification for a telemetry interface unit (TIU) for connection to the PSTN', BT Document Reference 0106-0325-3-00002 (CMC).
- 11 K E Nolde: 'Telemetry interface unit (TIU) Message Protocol (RITA protocol)', BT document reference 0106-0325-3-00004 (CMC).



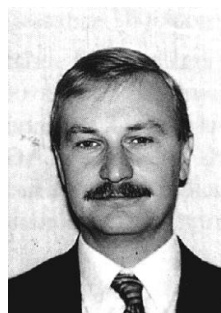
Kevin Welsby joined BT in 1967 as an apprentice. Having received a BSc in Electrical Engineering from Aston University in 1978, he joined BT Laboratories to work on the development of telephone testing standards and techniques. Later he moved to the development of telephone instruments

He started design and specification work for telemetry in 1992, and now leads the team delivering the network systems (telemetry platform and interface unit).



Jeremy Adams received a joint honours degree in Philosophy and Psychology from Bristol University in 1970 and joined BT Laboratories in 1971. He spent nine years as BTL's technical editor (during which time he produced the first four issues of this journal) and then worked on technical documentation for the Westminster switched star cable TV system, before becoming involved in local loop architecture evolution studies.

He has been involved in telemetry since 1990 and currently works closely with BT's telemetry product line as their trials manager.



Jeff Deslandes joined BT in 1973 as an apprentice. He moved to BT Laboratories in 1982 after gaining a Bachelors degree at Liverpool Polytechnic. He went on to gain a Masters degree in 1987 and a PhD in 1991, both from the University of Essex. He has worked on analogue line interfaces for digital exchanges and digital transmission techniques. Since 1992 he has worked on solutions designs and is the system design authority for the telemetry system overall. He is a Chartered Engineer and Member of the IEE.